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ADMINISTRATION RESEAUX

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TP: GESTION DES UTILISATEURS LINUX

Introduction :

Linux, en tant que système open source, offre une personnalisation approfondie et une sécurité renforcée grâce à son architecture modulaire.

Contrairement aux systèmes propriétaires, Linux permet aux administrateurs de contrôler finement les droits d'accès et les politiques de sécurité.

Ce TP explore la gestion des utilisateurs, le shadowing des mots de passe, et les bonnes pratiques pour sécuriser un système Linux.

I. Exploration de l'arborescence

- Afficher l'arborescence linux:

>ls/

- Voir le chemin de quelques commandes:

>which ls



II. Shadowing des mots de passe

- **Problématique** :

Le fichier /etc/passwd stocke historiquement les mots de passe hachés, mais ceux-ci sont accessibles en lecture à tous les utilisateurs (*others*). Cela expose les hachages à des attaques par dictionnaire.

- **Solution** :

Le **shadowing** déplace les hachages vers /etc/shadow, accessible uniquement par root.

1. Désactiver le shadowing

>pwunconv

>cat /etc/passwd

```
user@ubuntu:~$ sudo pwunconv
user@ubuntu:~$ cat /etc/passwd
root:$6$uVBJAdwo$yefOR.esZqlXfAm/82Ya69FKM5RTCrApS02I3vRlV3GsWHpjpiWPpUz8tJoWkOWcIZO6b6ONLs01JxFOTU3jc.:0:0:root:/root:/bin/bash
daemon:*:1:1:daemon:/usr/sbin:/bin/sh
bin:*:2:2:bin:/bin:/bin/sh
sys:*:3:3:sys:/dev:/bin/sh
sync:*:4:65534:sync:/bin:/bin/sync
games:*:5:60:games:/usr/games:/bin/sh
man:*:6:12:man:/var/cache/man:/bin/sh
lp:*:7:7:lp:/var/spool/lpd:/bin/sh
mail:*:8:8:mail:/var/mail:/bin/sh
news:*:9:9:news:/var/spool/news:/bin/sh
uucp:*:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:*:13:13:proxy:/bin:/bin/sh
www-data:*:33:33:www-data:/var/www:/bin/sh
backup:*:34:34:backup:/var/backups:/bin/sh
list:*:38:38:Mailing List Manager:/var/list:/bin/sh
irc:*:39:39:ircd:/var/run/ircd:/bin/sh
gnats:*:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:*:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:!:100:101::/var/lib/libuuid:/bin/sh
syslog:*:101:103::/home/syslog:/bin/false
messagebus:*:102:107::/var/run/dbus:/bin/false
avahi-autoipd:*:103:110:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/bin/false
avahi:*:104:111:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/bin/false
couchdb:*:105:113:CouchDB Administrator,,,:/var/lib/couchdb:/bin/bash
speech-dispatcher:!:106:29:Speech Dispatcher,,,:/var/run/speech-dispatcher:/bin/sh
usbmux:*:107:46:usbmux daemon,,,:/home/usbmux:/bin/false
haldaemon:*:108:114:Hardware abstraction layer,,,:/var/run/hald:/bin/false
kernoops:*:109:65534:Kernel Oops Tracking Daemon,,,:/:/bin/false
pulse:*:110:115:PulseAudio daemon,,,:/var/run/pulse:/bin/false
rtkit:*:111:117:RealtimeKit,,,:/proc:/bin/false
saned:*:112:118::/home/saned:/bin/false
hplip:*:113:7:HPLIP system user,,,:/var/run/hplip:/bin/false
gdm:*:114:120:Gnome Display Manager:/var/lib/gdm:/bin/false
user:$1$kbjo9m5L$7yO2YcFgyFnM5BSrFPa.H1:1000:1000:ubuntu1.10.04,,,:/home/user:/bin/bash
```

1) Le mot de passe est haché 1kbjo9m5L$7yO2YcFgyFnM5BSrFPa.H1 avec l'algorithme MD5.
2) La valeur du sel est: $kbjo9m5L, c'est un élément clé pour sécuriser les mots de passe.
3) Il rend les attaques massives impraticables et garantit qu'un même mot de passe **ne produira jamais** le même hachage sur un système. Son utilisation est indispensable dans toute application ou système moderne.

>ls –l /etc/passwd

```
user@ubuntu:~$ ls -l /etc/passwd
-rw-r--r-- 1 root root 2661 2025-04-22 04:30 /etc/passwd
```

1) Permissions :
    - Utilisateur : lecture et modification
    - Groupe : lecture seule
    - Autres : lecture seule

2) Oui, les utilisateurs "Autres" peuvent visualiser le mot de passe chiffré, car ils disposent d'un accès en lecture.

3) Le risque, c'est que cela peut faciliter l'identification ou la récupération du mot de passe, surtout si le chiffrement est faible ou connu. Il pourrait être possible de le déchiffrer d'une autre manière.

2. Activer le shadowing

>pwunconv

>cat /etc/passwd

```
user@ubuntu:~$ sudo pwconv
user@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
syslog:x:101:103::/home/syslog:/bin/false
messagebus:x:102:107::/var/run/dbus:/bin/false
avahi-autoipd:x:103:110:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/bin/false
avahi:x:104:111:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/bin/false
couchdb:x:105:113:CouchDB Administrator,,,:/var/lib/couchdb:/bin/bash
```

→ On remarque que les mots de passe ne sont pas hachés.

>cat /etc/shadow

```
user@ubuntu:~$ cat /etc/shadow
cat: /etc/shadow: Permission denied
```

>ls -l /etc/shadow

```
user@ubuntu:~$ ls -l /etc/shadow
-r--r----- 1 root shadow 1750 2025-04-22 04:33 /etc/shadow
```

→ Le fichier n'est plus accessible par les Others.

III. Utilisateurs “humain” et “systems” :

>/etc/passwd

```
user@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
syslog:x:101:103::/home/syslog:/bin/false
messagebus:x:102:107::/var/run/dbus:/bin/false
avahi-autoipd:x:103:110:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/bin/false
avahi:x:104:111:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/bin/false
couchdb:x:105:113:CouchDB Administrator,,,:/var/lib/couchdb:/bin/bash
speech-dispatcher:x:106:29:Speech Dispatcher,,,:/var/run/speech-dispatcher:/bin/sh
usbmux:x:107:46:usbmux daemon,,,:/home/usbmux:/bin/false
haldaemon:x:108:114:Hardware abstraction layer,,,:/var/run/hald:/bin/false
kernoops:x:109:65534:Kernel Oops Tracking Daemon,,,:/:/bin/false
pulse:x:110:115:PulseAudio daemon,,,:/var/run/pulse:/bin/false
rtkit:x:111:117:RealtimeKit,,,:/proc:/bin/false
saned:x:112:118::/home/saned:/bin/false
hplip:x:113:7:HPLIP system user,,,:/var/run/hplip:/bin/false
gdm:x:114:120:Gnome Display Manager:/var/lib/gdm:/bin/false
user:x:1000:1000:ubuntu1.10.04,,,:/home/user:/bin/bash
snmp:x:115:123::/var/lib/snmp:/bin/false
sshd:x:116:65534::/var/run/sshd:/usr/sbin/nologin
user8:x:1001:1001::/home/user8:/bin/sh
user30:x:1002:1002::/home/user30:/bin/sh
user3:x:1003:1005::/home/user3:/bin/sh
user4:x:1004:1006::/home/user4:/bin/sh
jack:x:1005:1008::/home/jack:/bin/sh
jaqueline:x:1006:1009::/home/jaqueline:/bin/sh
```

→ Exemple des utilisateurs humains : user8 – user30 – user4.

→ Exemple des utilisateurs systèmes : root – daemon – man .

IV. Ajout et suppression de compte :

1. Ajout de compte via ligne de commande:

Une seule commande:

>useradd –m user3 –p user3

```
user@ubuntu:~$ useradd -m user3 -p user3
useradd: user 'user3' already exists
```

>ls /home

Pour verifier que user3 existe:

```
user@ubuntu:~$ ls /home
jack  jaqueline  user  user3  user8
```

La meme commande sans –m:

```
user@ubuntu:~$ sudo useradd user9 -p user9
user@ubuntu:~$ ls /home
jack  jaqueline  user  user3  user8
```

→ On Remarque que user9 n'est pas ajouté

En deux temps:

>useradd –m user3

>passwd user3

```
user@ubuntu:~$ sudo useradd user2
user@ubuntu:~$ sudo passwd user2
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
```

Avec GECOS:

```
root@ubuntu:/home/user# adduser utilisateur4
Adding user `utilisateur4' ...
Adding new group `utilisateur4' (1018) ...
Adding new user `utilisateur4' (1004) with group `utilisateur4' ...
Creating home directory `/home/utilisateur4' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for utilisateur4
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
root@ubuntu:/home/user# exit
exit
```

2. Consultation du dernier changement du mot de passe:

>sudo cat /etc/shadow | grep user3

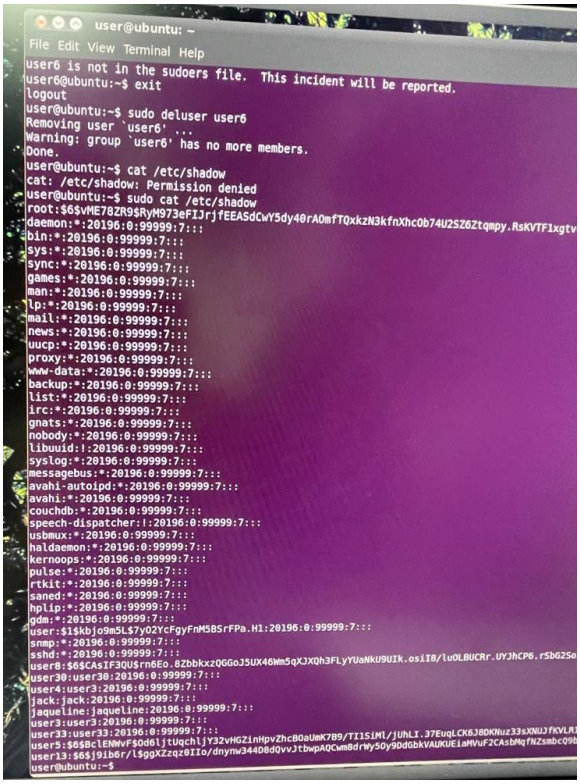
```
user@ubuntu:~$ sudo pwconv
user@ubuntu:~$ sudo cat /etc/shadow | grep utilisateur3
utilisateur3:utilisateur3:20200:0:99999:7:::
```

>date –d "1970-01-01 + val1 days"

```
user@ubuntu:~$ date -d "1970-01-01 + 7 days"
Thu Jan  8 00:00:00 PST 1970
```

3. Suppression de comptes:

>userdel user1



4. UID d'un user:

>id –u user3

```
user@ubuntu:~$ id -u utilisateur4
1004
```

5. Afficher le groupe principal d'appartenance de l'utilisateur:

>id –gn user3

```
user@ubuntu:~$ id -gn utilisateur4
utilisateur4
```

6. Liste des noms de tous les groups auxquels appartient l'utlisateur:

>id –Gn user3

```
user@ubuntu:~$ id -Gn utilisateur4
utilisateur4
```

L'option ***–aG*** est utilisée pour ajouter l'utilisateur à un groupe supplémentaire sans le retirer de ses groupes actuels.

```
user@ubuntu:~$ sudo groupadd devgroup
user@ubuntu:~$ sudo usermod -aG devgroup utilisateur4
user@ubuntu:~$ id -Gn utilisateur4
utilisateur4 devgroup
user@ubuntu:~$
```

V. Switching users :

Activer le compte root :

>passwd

```
user@ubuntu:~$ sudo su
root@ubuntu:/home/user# passwd root
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
```

Switching users vers le root:

>su –

>su root

```
root@ubuntu:/home/user# su -
root@ubuntu:~# su root
root@ubuntu:~# exit
exit
root@ubuntu:~# exit
logout
root@ubuntu:/home/user# exit
exit
```

Switching users entre les autres comptes:

>su –s /bin/bash – user3

>pwd

```
user@ubuntu:~$ su -s /bin/bash - utilisateur4
Password:
```

>exit

>su –s /bin/bash – user3

>pwd

```
utilisateur4@ubuntu:~$ su -s /bin/bash utilisateur4
Password:
```

VI. Sudoers:

```
user@ubuntu:~$ sudo adduser user88 sudo
Adding user `user88' to group `sudo' ...
Adding user user88 to group sudo
Done.
user@ubuntu:~$ sudo su -s /bin/bash - user88
user88@ubuntu:~$ sudo /etc/sahdow
[sudo] password for user88:
sudo: /etc/sahdow: command not found
user88@ubuntu:~$ sudo /etc/shadow
sudo: /etc/shadow: command not found
user88@ubuntu:~$ sudo cat /etc/shadow
root:$6$8bGzAMOr$MRL5PP0ZczFFf.8C8IDIWH51wa6MpUdoe/OYHX6zBRWJnxzZtrX1np1RB4RjQx6.aSjZUtMKtY1DGANMfzCd31:20200:0:99999:7:::
daemon:*:20200:0:99999:7:::
bin:*:20200:0:99999:7:::
sys:*:20200:0:99999:7:::
sync:*:20200:0:99999:7:::
games:*:20200:0:99999:7:::
man:*:20200:0:99999:7:::
lp:*:20200:0:99999:7:::
```

VII. Propriétés d'un compte : « change age »

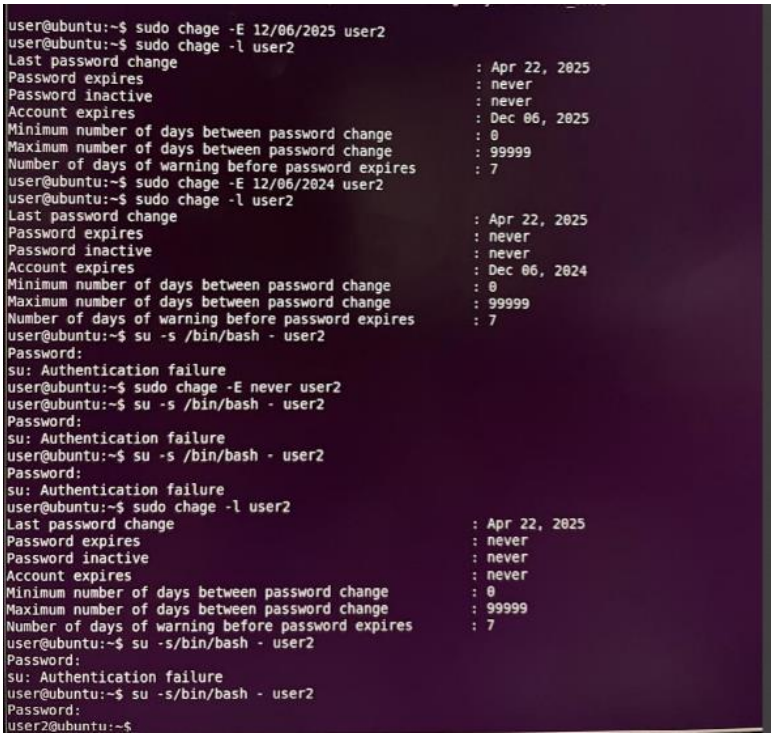
Date du changement de password:

```
user@ubuntu:~$ sudo chage -l user2
chage: user 'user2' does not exist in /etc/passwd
user@ubuntu:~$ sudo chage -l user4
Last password change                                    : Apr 18, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ sudo passwd user4
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
user@ubuntu:~$ sudo chage -l user4
Last password change                                    : Apr 18, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$
```

Verrouillage d'un compte:



Expiration d’un compte:



```
user@ubuntu:~$ sudo chage -E 12/06/2025 user2
user@ubuntu:~$ sudo chage -l user2
Last password change                                    : Apr 22, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : Dec 06, 2025
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ sudo chage -E 12/06/2024 user2
user@ubuntu:~$ sudo chage -l user2
Last password change                                    : Apr 22, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : Dec 06, 2024
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ su -s /bin/bash - user2
Password:
su: Authentication failure
user@ubuntu:~$ sudo chage -E never user2
user@ubuntu:~$ su -s /bin/bash - user2
Password:
su: Authentication failure
user@ubuntu:~$ su -s /bin/bash - user2
Password:
su: Authentication failure
user@ubuntu:~$ sudo chage -l user2
Last password change                                    : Apr 22, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ su -s/bin/bash - user2
Password:
su: Authentication failure
user@ubuntu:~$ su -s/bin/bash - user2
Password:
user2@ubuntu:~$
```

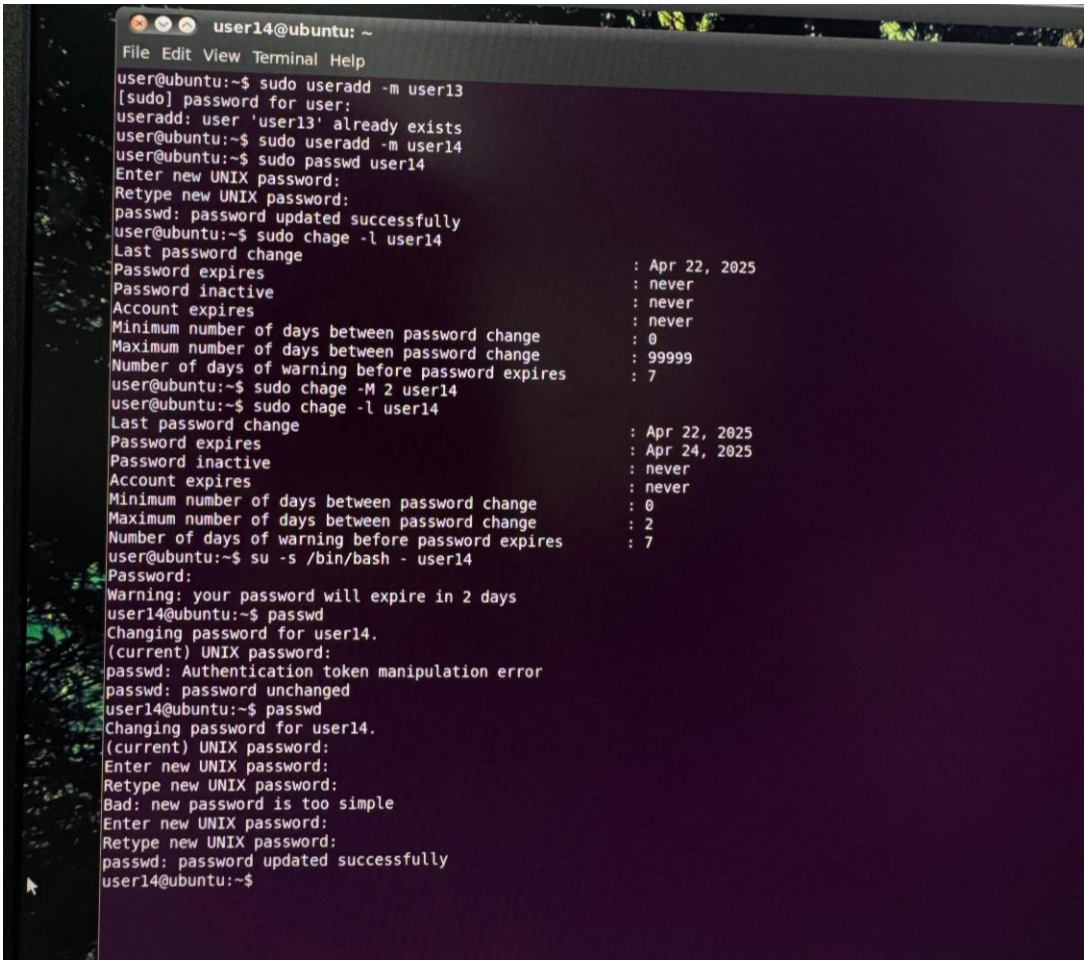
```
user@ubuntu:~$ sudo chage -l user2
Last password change                                    : Apr 18, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ sudo chage -E never user2
user@ubuntu:~$ sudo chage -E 04/01/2024
Usage: chage [options] [LOGIN]

Options:
  -d, --lastday LAST_DAY        set date of last password change to LAST_DAY
  -E, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -h, --help                    display this help message and exit
  -I, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -l, --list                    show account aging information
  -m, --mindays MIN_DAYS        set minimum number of days before password
                                change to MIN_DAYS
  -M, --maxdays MAX_DAYS        set maximim number of days before password
                                change to MAX_DAYS
  -W, --warndays WARN_DAYS      set expiration warning days to WARN_DAYS

user@ubuntu:~$ sudo chage -l user2
Last password change                                    : Apr 18, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ su -s /bin/bash - user2
Password:
su: Authentication failure
```

Périodicité de Mise à jour du mot de passe :

```
user@ubuntu:~$ sudo useradd -m user13
user@ubuntu:~$ sudo passwd user13
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
user@ubuntu:~$ sudo chage -l user13
Last password change                                    : Apr 18, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
user@ubuntu:~$ sudo chage -M 2 user13
user@ubuntu:~$ sudo chage -l user13
Last password change                                    : Apr 18, 2025
Password expires                                        : Apr 20, 2025
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 2
Number of days of warning before password expires       : 7
user@ubuntu:~$ su -s /bin/bash - user13
Password:
su: Authentication failure
user@ubuntu:~$ su -s /bin/bash - user13
Password:
Warning: your password will expire in 2 days
user13@ubuntu:~$ sudo passwd
[sudo] password for user13:
user13 is not in the sudoers file.  This incident will be reported.
```



Conclusion:

Dans ce TP, nous avons développé plusieurs compétences importantes liées à l'administration système.

Nous avons appris à ouvrir et configurer une machine virtuelle, à gérer le système de *shadowing* pour la protection des mots de passe, ainsi qu'à manipuler les fichiers de mots de passe et à appliquer différentes méthodes de sécurisation.

Nous avons également acquis de l'expérience dans la création d'utilisateurs, la gestion des comptes système et des comptes utilisateurs humains.

Enfin, nous avons vu comment modifier les propriétés d'un compte de différentes façons, en fonction des besoins spécifiques.

TP: Compactage et Archivage

COMPACTAGE :

Création d'un fichier :

>ls –l new1

Taille du fichier :

>ls –l new1

>ls –lh new1

Compactage :

>gzip new1

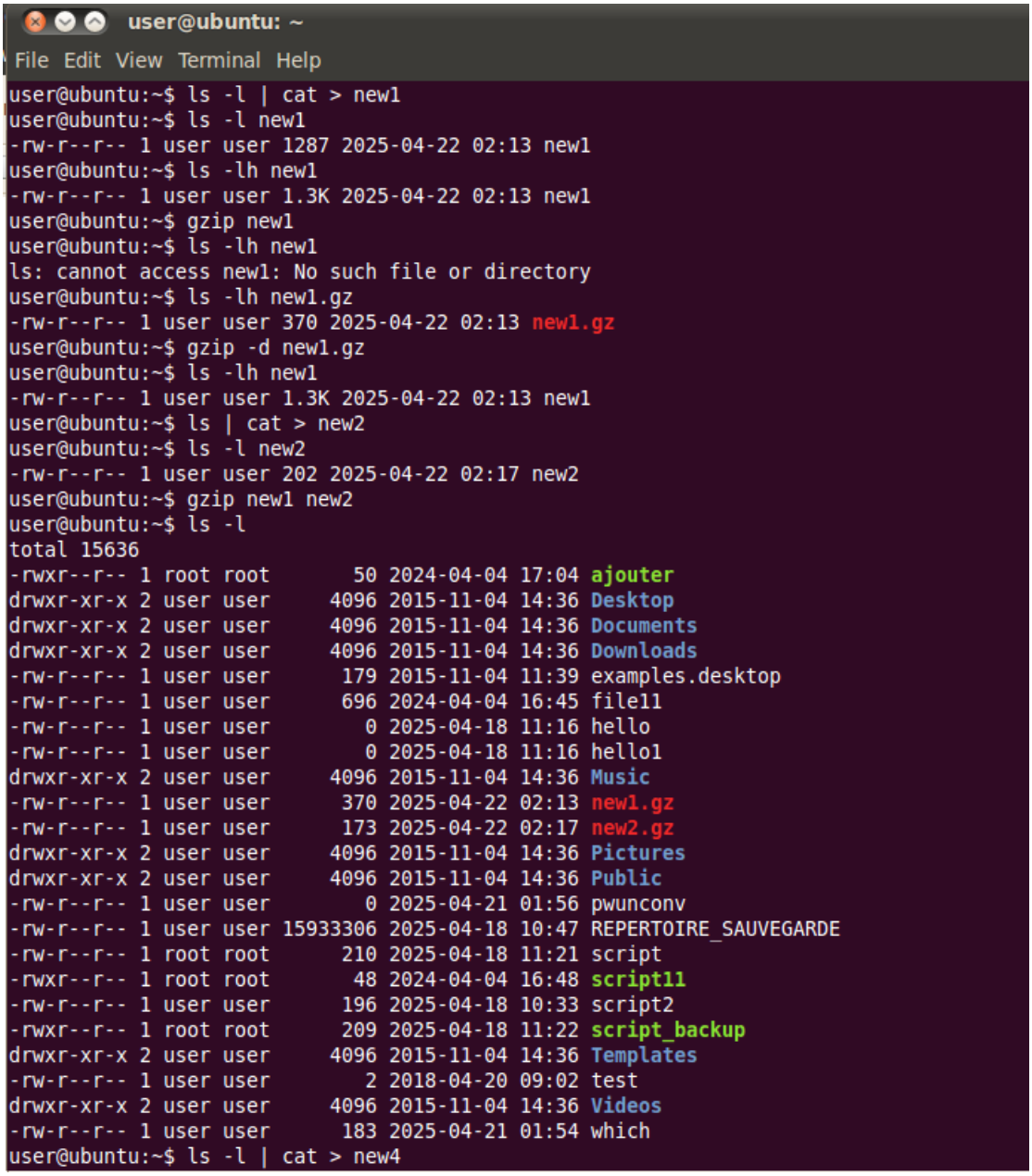
>ls –lh new1

>ls –lh new1.gz

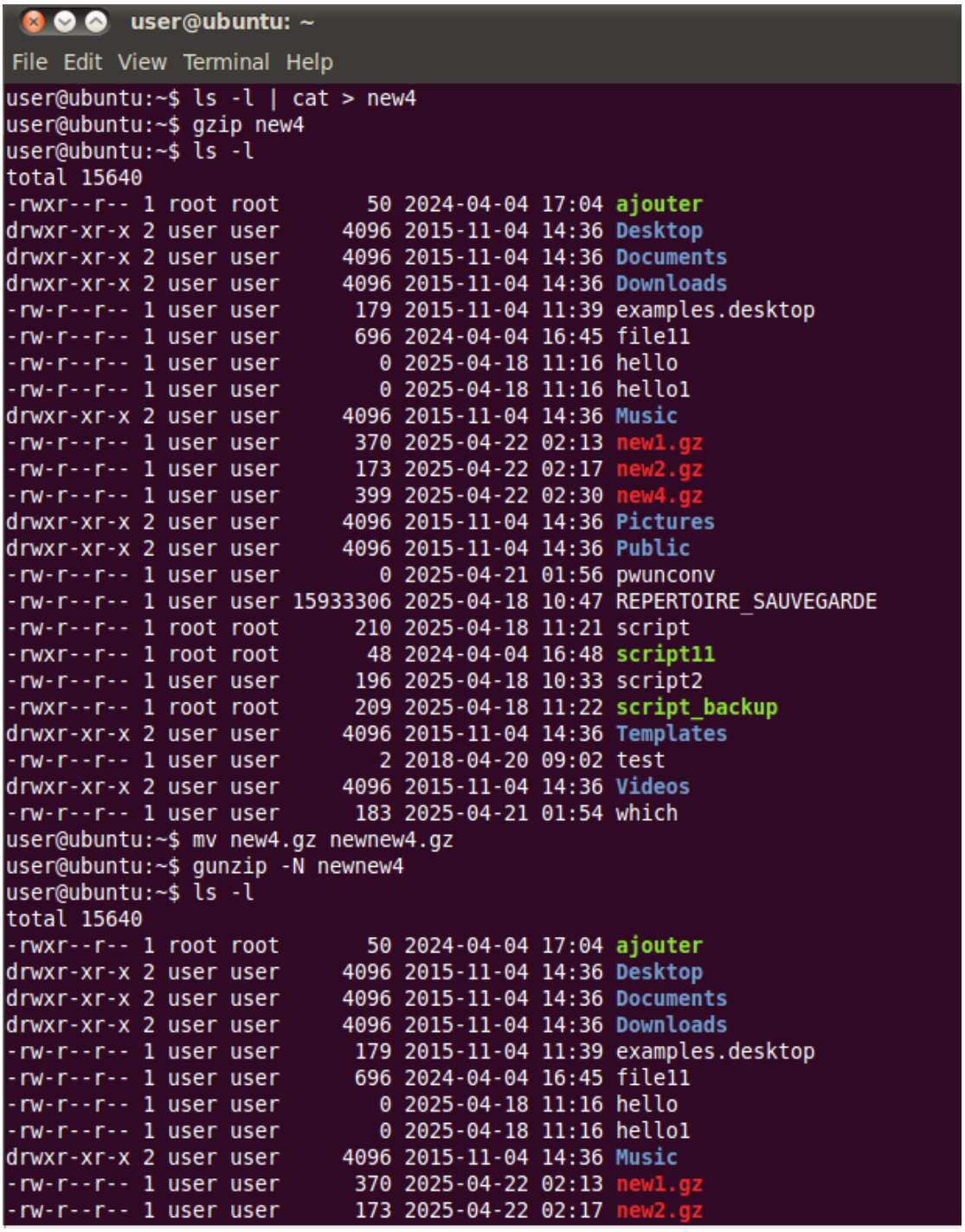
Décompactage :

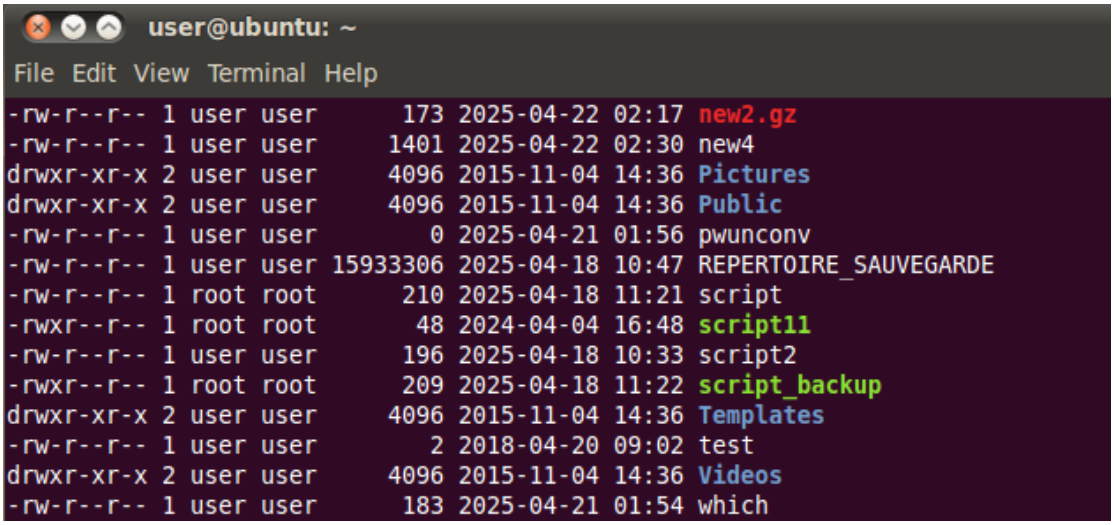
>gzip -d new1.gzip

>ls –lh new1


user@ubuntu: ~
File Edit View Terminal Help
user@ubuntu:~$ ls -l | cat > new1
user@ubuntu:~$ ls -l new1
-rw-r--r-- 1 user user 1287 2025-04-22 02:13 new1
user@ubuntu:~$ ls -lh new1
-rw-r--r-- 1 user user 1.3K 2025-04-22 02:13 new1
user@ubuntu:~$ gzip new1
user@ubuntu:~$ ls -lh new1
ls: cannot access new1: No such file or directory
user@ubuntu:~$ ls -lh new1.gz
-rw-r--r-- 1 user user 370 2025-04-22 02:13 new1.gz
user@ubuntu:~$ gzip -d new1.gz
user@ubuntu:~$ ls -lh new1
-rw-r--r-- 1 user user 1.3K 2025-04-22 02:13 new1
user@ubuntu:~$ ls | cat > new2
user@ubuntu:~$ ls -l new2
-rw-r--r-- 1 user user 202 2025-04-22 02:17 new2
user@ubuntu:~$ gzip new1 new2
user@ubuntu:~$ ls -l
total 15636
-rwxr--r-- 1 root root 50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user 179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user 696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user 370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user 173 2025-04-22 02:17 new2.gz
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user 0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root 210 2025-04-18 11:21 script
-rwxr--r-- 1 root root 48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user 196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root 209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user 2 2018-04-20 09:02 test
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user 183 2025-04-21 01:54 which
user@ubuntu:~$ ls -l | cat > new4

Scénario1 :


user@ubuntu: ~
File Edit View Terminal Help
user@ubuntu:~$ ls -l | cat > new4
user@ubuntu:~$ gzip new4
user@ubuntu:~$ ls -l
total 15640
-rwxr--r-- 1 root root 50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user 179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user 696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user 370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user 173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user 399 2025-04-22 02:30 new4.gz
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user 0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root 210 2025-04-18 11:21 script
-rwxr--r-- 1 root root 48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user 196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root 209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user 2 2018-04-20 09:02 test
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user 183 2025-04-21 01:54 which
user@ubuntu:~$ mv new4.gz newnew4.gz
user@ubuntu:~$ gunzip -N newnew4
user@ubuntu:~$ ls -l
total 15640
-rwxr--r-- 1 root root 50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user 179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user 696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user 370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user 173 2025-04-22 02:17 new2.gz


user@ubuntu: ~
File Edit View Terminal Help
-rw-r--r-- 1 user user 173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user 1401 2025-04-22 02:30 new4
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user 0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root 210 2025-04-18 11:21 script
-rwxr--r-- 1 root root 48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user 196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root 209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user 2 2018-04-20 09:02 test
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user 183 2025-04-21 01:54 which

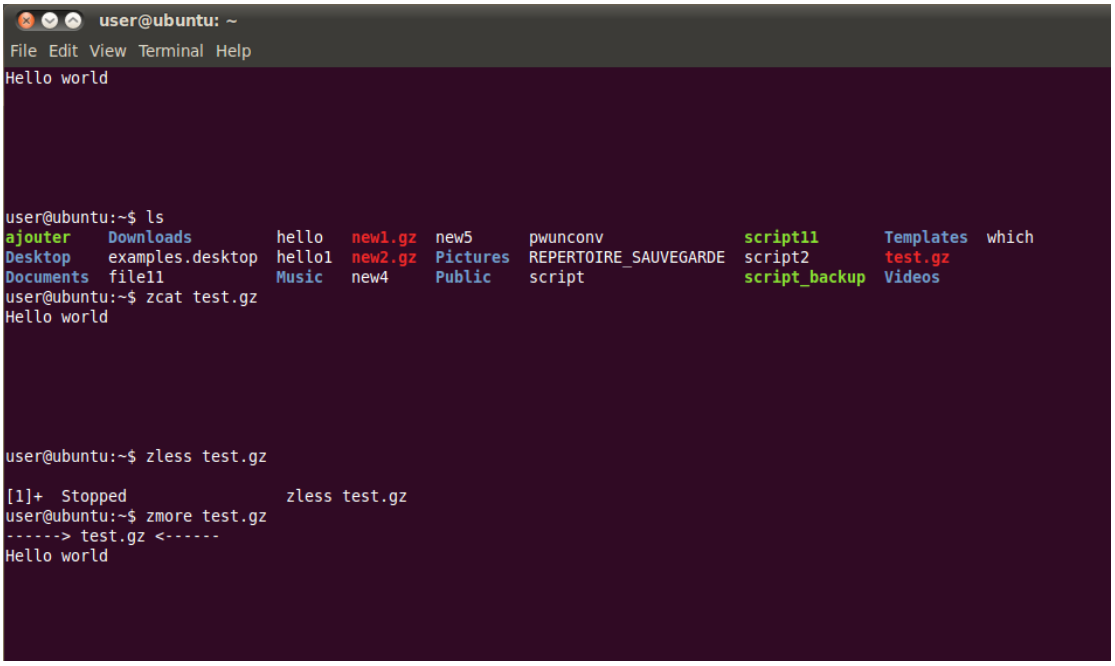
Scénario2 :

```
user@ubuntu:~$ ls -l | cat > new5
user@ubuntu:~$ gzip new5
user@ubuntu:~$ ls -l
total 15644
-rwxr--r-- 1 root root       50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user      179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user      696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user      370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user      173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user     1401 2025-04-22 02:30 new4
-rw-r--r-- 1 user user      406 2025-04-22 02:39 new5.gz
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user        0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root      210 2025-04-18 11:21 script
-rwxr--r-- 1 root root       48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user      196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root      209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user        2 2018-04-20 09:02 test
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user      183 2025-04-21 01:54 which
```

```
user@ubuntu: ~
File  Edit  View  Terminal  Help
user@ubuntu:~$ mv new5.gz newnew5.gz
user@ubuntu:~$ gunzip -N newnew5
user@ubuntu:~$ ls -l
total 15644
-rwxr--r-- 1 root root       50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user      179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user      696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user      370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user      173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user     1401 2025-04-22 02:30 new4
-rw-r--r-- 1 user user     1455 2025-04-22 02:39 new5
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user        0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root      210 2025-04-18 11:21 script
-rwxr--r-- 1 root root       48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user      196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root      209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user        2 2018-04-20 09:02 test
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user      183 2025-04-21 01:54 which
```

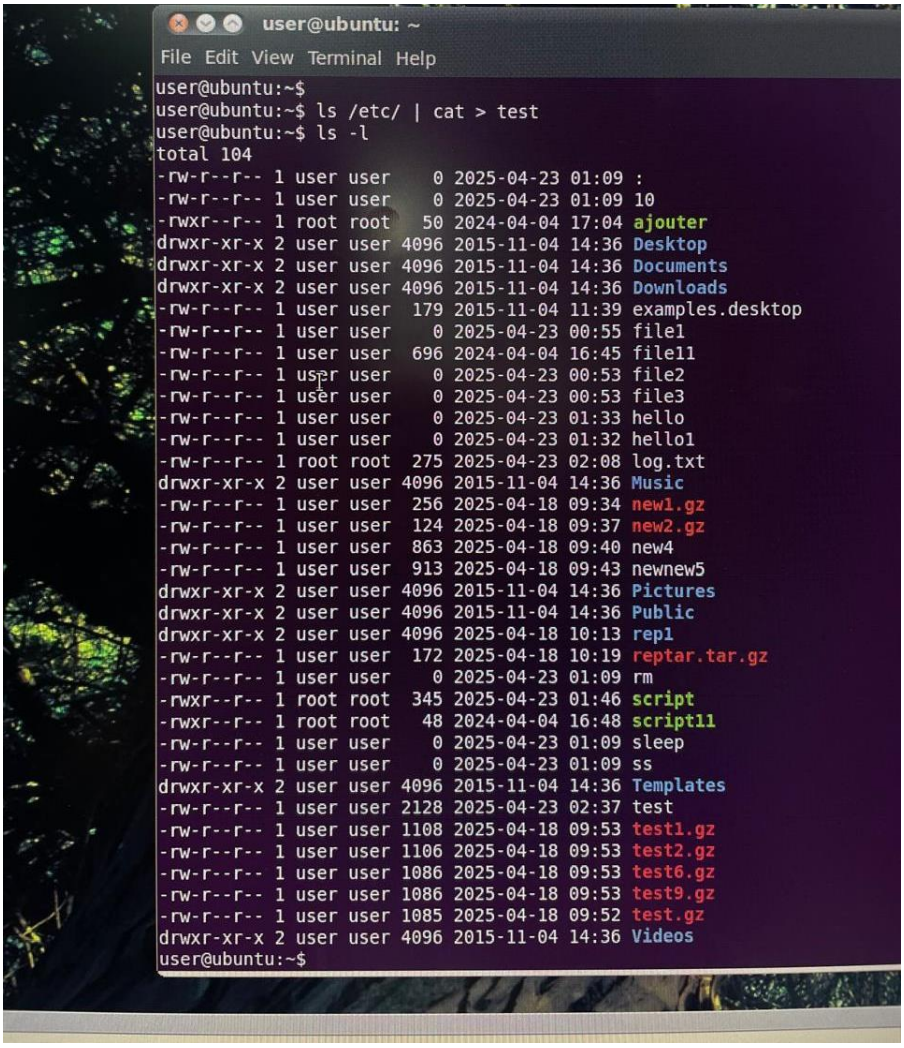
Décompactage vers l'écran :





Qualité de compactage :

Création du fichier :



Duplication du fichier :

```
user@ubuntu:~$ cp test test1
user@ubuntu:~$ cp test test2
user@ubuntu:~$ cp test test6
user@ubuntu:~$ cp test test9
user@ubuntu:~$ ls -lh
total 16M
-rwxr--r-- 1 root root   50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user  179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user  696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user    0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user    0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Music
-rw-r--r-- 1 user user  370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user  173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user 1.4K 2025-04-22 02:30 new4
-rw-r--r-- 1 user user 1.5K 2025-04-22 02:39 new5
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Public
-rw-r--r-- 1 user user    0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user  16M 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root  210 2025-04-18 11:21 script
```

```
user@ubuntu: ~
File  Edit  View  Terminal  Help
-rw-r--r-- 1 root root  210 2025-04-18 11:21 script
-rwxr--r-- 1 root root   48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user  196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root  209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user 2.1K 2025-04-22 03:00 test
-rw-r--r-- 1 user user 2.1K 2025-04-22 03:02 test1
-rw-r--r-- 1 user user 2.1K 2025-04-22 03:02 test2
-rw-r--r-- 1 user user 2.1K 2025-04-22 03:02 test6
-rw-r--r-- 1 user user 2.1K 2025-04-22 03:03 test9
-rw-r--r-- 1 user user   38 2025-04-22 02:42 test.gz
drwxr-xr-x 2 user user 4.0K 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user  183 2025-04-21 01:54 which
```

Compactage :

```
user@ubuntu:~$ gzip -l test1

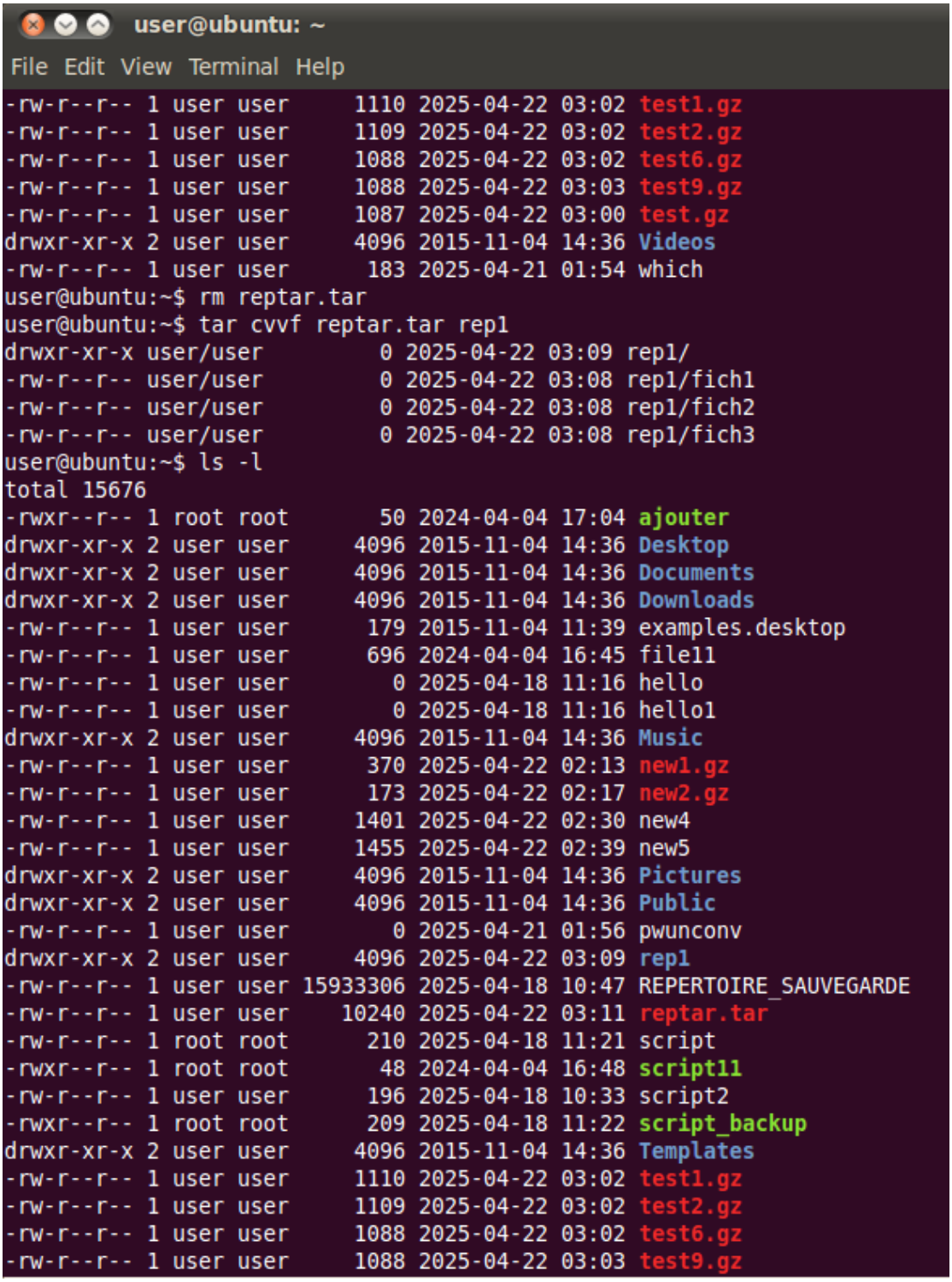
gzip: test1: not in gzip format
user@ubuntu:~$ gzip -1 test1
user@ubuntu:~$ gzip -2 test2
user@ubuntu:~$ gzip -6 test6
user@ubuntu:~$ gzip -9 test9
user@ubuntu:~$ gzip test
gzip: test.gz already exists; do you wish to overwrite (y or n)? y
user@ubuntu:~$ ls -l
total 15660
-rwxr--r-- 1 root root       50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user      179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user      696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user      370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user      173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user     1401 2025-04-22 02:30 new4
-rw-r--r-- 1 user user     1455 2025-04-22 02:39 new5
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user        0 2025-04-21 01:56 pwunconv
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 root root      210 2025-04-18 11:21 script
-rwxr--r-- 1 root root       48 2024-04-04 16:48 script11
```

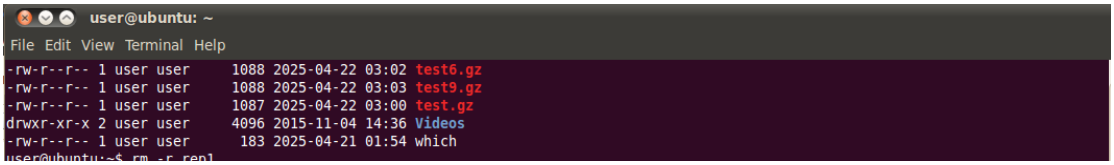
```
user@ubuntu: ~
File  Edit  View  Terminal  Help
-rwxr--r-- 1 root root       48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user      196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root      209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user     1110 2025-04-22 03:02 test1.gz
-rw-r--r-- 1 user user     1109 2025-04-22 03:02 test2.gz
-rw-r--r-- 1 user user     1088 2025-04-22 03:02 test6.gz
-rw-r--r-- 1 user user     1088 2025-04-22 03:03 test9.gz
-rw-r--r-- 1 user user     1087 2025-04-22 03:00 test.gz
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user      183 2025-04-21 01:54 which
```

ARCHIVAGE :

Archivage :

```
user@ubuntu:~$ touch fich1 fich2 fich3
user@ubuntu:~$ mkdir rep1
user@ubuntu:~$ mv fich1 fich2 fich3 rep1
user@ubuntu:~$ ls rep1
fich1  fich2  fich3
user@ubuntu:~$ tar cf reptar.tar rep1
user@ubuntu:~$ ls -l
total 15676
-rwxr--r-- 1 root root       50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user      179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user      696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user        0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user      370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user      173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user     1401 2025-04-22 02:30 new4
-rw-r--r-- 1 user user     1455 2025-04-22 02:39 new5
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user        0 2025-04-21 01:56 pwunconv
drwxr-xr-x 2 user user     4096 2025-04-22 03:09 rep1
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 user user    10240 2025-04-22 03:11 reptar.tar
-rw-r--r-- 1 root root      210 2025-04-18 11:21 script
-rwxr--r-- 1 root root       48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user      196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root      209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user     4096 2015-11-04 14:36 Templates
```


user@ubuntu: ~
File Edit View Terminal Help
-rw-r--r-- 1 user user 1110 2025-04-22 03:02 test1.gz
-rw-r--r-- 1 user user 1109 2025-04-22 03:02 test2.gz
-rw-r--r-- 1 user user 1088 2025-04-22 03:02 test6.gz
-rw-r--r-- 1 user user 1088 2025-04-22 03:03 test9.gz
-rw-r--r-- 1 user user 1087 2025-04-22 03:00 test.gz
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user 183 2025-04-21 01:54 which
user@ubuntu:~$ rm reptar.tar
user@ubuntu:~$ tar cvvf reptar.tar rep1
drwxr-xr-x user/user 0 2025-04-22 03:09 rep1/
-rw-r--r-- user/user 0 2025-04-22 03:08 rep1/fich1
-rw-r--r-- user/user 0 2025-04-22 03:08 rep1/fich2
-rw-r--r-- user/user 0 2025-04-22 03:08 rep1/fich3
user@ubuntu:~$ ls -l
total 15676
-rwxr--r-- 1 root root 50 2024-04-04 17:04 ajouter
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Desktop
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Documents
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Downloads
-rw-r--r-- 1 user user 179 2015-11-04 11:39 examples.desktop
-rw-r--r-- 1 user user 696 2024-04-04 16:45 file11
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello
-rw-r--r-- 1 user user 0 2025-04-18 11:16 hello1
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Music
-rw-r--r-- 1 user user 370 2025-04-22 02:13 new1.gz
-rw-r--r-- 1 user user 173 2025-04-22 02:17 new2.gz
-rw-r--r-- 1 user user 1401 2025-04-22 02:30 new4
-rw-r--r-- 1 user user 1455 2025-04-22 02:39 new5
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Pictures
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Public
-rw-r--r-- 1 user user 0 2025-04-21 01:56 pwunconv
drwxr-xr-x 2 user user 4096 2025-04-22 03:09 rep1
-rw-r--r-- 1 user user 15933306 2025-04-18 10:47 REPERTOIRE_SAUVEGARDE
-rw-r--r-- 1 user user 10240 2025-04-22 03:11 reptar.tar
-rw-r--r-- 1 root root 210 2025-04-18 11:21 script
-rwxr--r-- 1 root root 48 2024-04-04 16:48 script11
-rw-r--r-- 1 user user 196 2025-04-18 10:33 script2
-rwxr--r-- 1 root root 209 2025-04-18 11:22 script_backup
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Templates
-rw-r--r-- 1 user user 1110 2025-04-22 03:02 test1.gz
-rw-r--r-- 1 user user 1109 2025-04-22 03:02 test2.gz
-rw-r--r-- 1 user user 1088 2025-04-22 03:02 test6.gz
-rw-r--r-- 1 user user 1088 2025-04-22 03:03 test9.gz


user@ubuntu: ~
File Edit View Terminal Help
-rw-r--r-- 1 user user 1088 2025-04-22 03:02 test6.gz
-rw-r--r-- 1 user user 1088 2025-04-22 03:03 test9.gz
-rw-r--r-- 1 user user 1087 2025-04-22 03:00 test.gz
drwxr-xr-x 2 user user 4096 2015-11-04 14:36 Videos
-rw-r--r-- 1 user user 183 2025-04-21 01:54 which

Extraction de l’archivage :

```
user@ubuntu:~$ rm -r rep1
user@ubuntu:~$ tar xvf reptar.tar
rep1/
rep1/fich1
rep1/fich2
rep1/fich3
user@ubuntu:~$ ls rep1
fich1  fich2  fich3
```

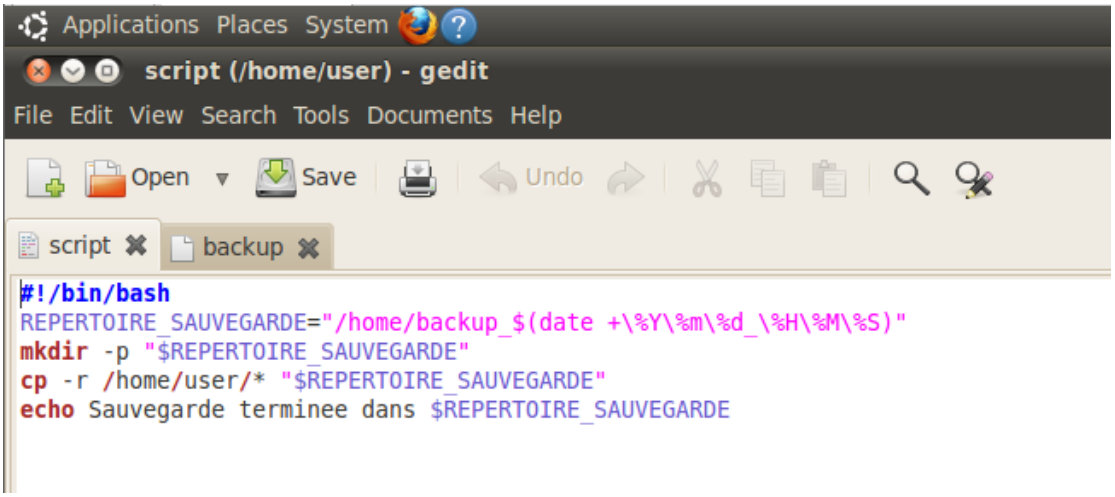
Archivage et compactage :

```
user@ubuntu:~$ touch fich1 fich2 fich3
user@ubuntu:~$ mkdir rep¹
user@ubuntu:~$ mv fich1 fich2 fich3 rep1
user@ubuntu:~$ ls rep1
fich1  fich2  fich3
user@ubuntu:~$ tar cvvf reptar.tar rep1
drwxr-xr-x user/user         0 2025-04-22 03:14 rep1/
-rw-r--r-- user/user         0 2025-04-22 03:13 rep1/fich1
-rw-r--r-- user/user         0 2025-04-22 03:13 rep1/fich2
-rw-r--r-- user/user         0 2025-04-22 03:13 rep1/fich3
user@ubuntu:~$ gzip reptar.tar
user@ubuntu:~$ ls
ajouter    Downloads         hello   new1.gz  new5      pwunconv  REPERTOIRE_SAUVEGARDE  script11       Templates  test6.gz  Videos
Desktop    examples.desktop  hello1  new2.gz  Pictures  rep1      reptar.tar.gz          script2        test1.gz   test9.gz  which
Documents  file11            Music   new4     Public    rep¹      script                 script_backup  test2.gz   test.gz
```

1ere méthode :

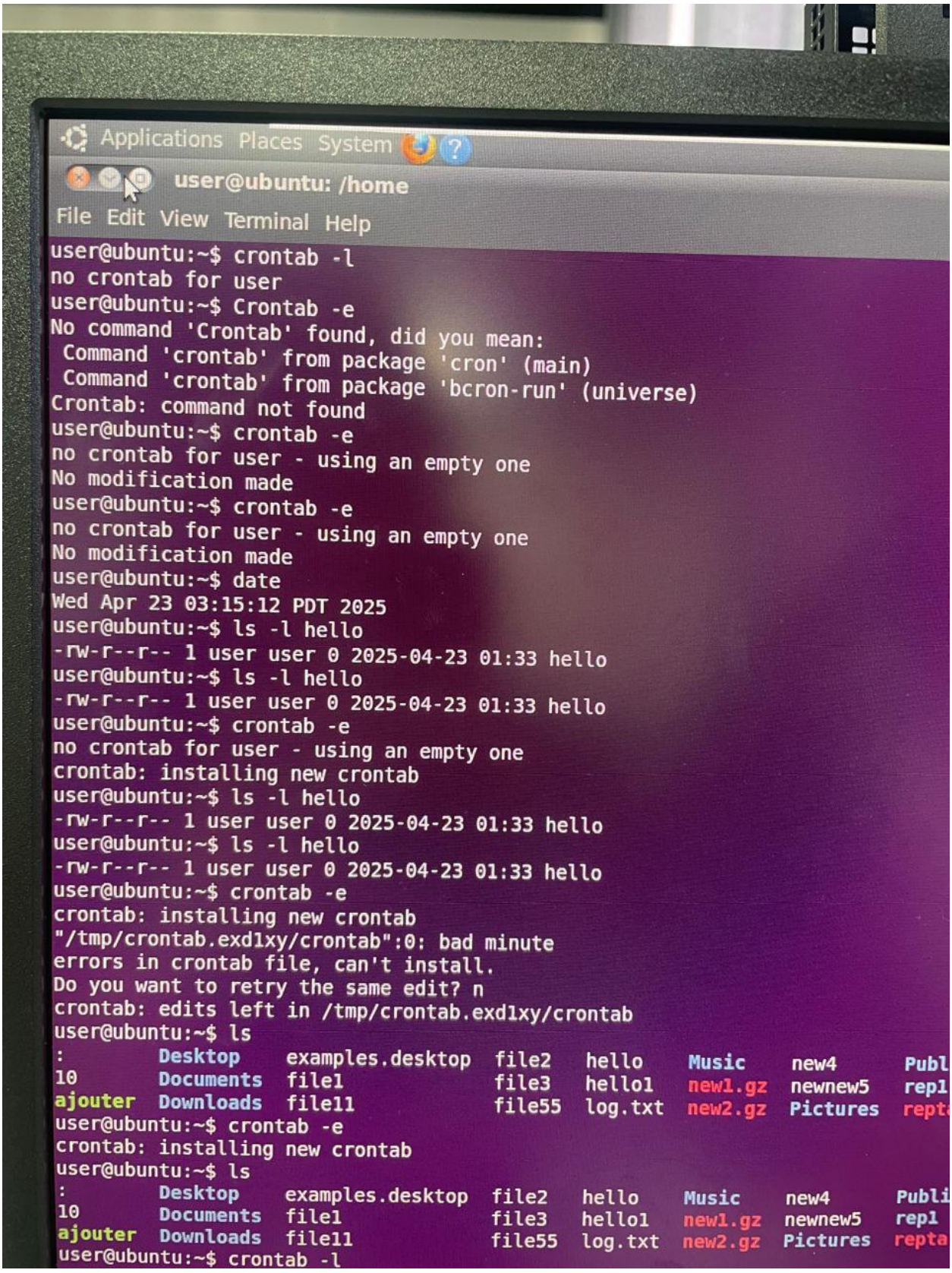
```
user@ubuntu:~$ gzip -d reptar.tar.gz
user@ubuntu:~$ ls
ajouter    Downloads         hello   new1.gz  new5      pwunconv  REPERTOIRE_SAUVEGARDE  script11       Templates  test6.gz  Videos
Desktop    examples.desktop  hello1  new2.gz  Pictures  rep1      reptar.tar             script2        test1.gz   test9.gz  which
Documents  file11            Music   new4     Public    rep¹      script                 script_backup  test2.gz   test.gz
user@ubuntu:~$ tar xvf reptar.tar
rep1/
rep1/fich1
rep1/fich2
rep1/fich3
user@ubuntu:~$ ls
ajouter    Downloads         hello   new1.gz  new5      pwunconv  REPERTOIRE_SAUVEGARDE  script11       Templates  test6.gz  Videos
Desktop    examples.desktop  hello1  new2.gz  Pictures  rep1      reptar.tar             script2        test1.gz   test9.gz  which
Documents  file11            Music   new4     Public    rep¹      script                 script_backup  test2.gz   test.gz
user@ubuntu:~$ tar cvf reptar.tar rep1
```

Exercice :




```
/home/user/.gconf/desktop/gnome/applications/
/home/user/.gconf/desktop/gnome/applications/window_manager/
/home/user/.gconf/desktop/gnome/applications/window_manager/%gconf.xml
/home/user/.gconf/desktop/gnome/applications/%gconf.xml
/home/user/.gconf/desktop/gnome/%gconf.xml
tar: Exiting with failure status due to previous errors
Sauvegarde terminée dans /home/backup_20250423_105359
```

TP: CRONTAB


Applications Places System
user@ubuntu: /home
File Edit View Terminal Help
user@ubuntu:~$ crontab -l
no crontab for user
user@ubuntu:~$ Crontab -e
No command 'Crontab' found, did you mean:
Command 'crontab' from package 'cron' (main)
Command 'crontab' from package 'bcron-run' (universe)
Crontab: command not found
user@ubuntu:~$ crontab -e
no crontab for user - using an empty one
No modification made
user@ubuntu:~$ crontab -e
no crontab for user - using an empty one
No modification made
user@ubuntu:~$ date
Wed Apr 23 03:15:12 PDT 2025
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 01:33 hello
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 01:33 hello
user@ubuntu:~$ crontab -e
no crontab for user - using an empty one
crontab: installing new crontab
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 01:33 hello
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 01:33 hello
user@ubuntu:~$ crontab -e
crontab: installing new crontab
"/tmp/crontab.exd1xy/crontab":0: bad minute
errors in crontab file, can't install.
Do you want to retry the same edit? n
crontab: edits left in /tmp/crontab.exd1xy/crontab
user@ubuntu:~$ ls
: Desktop examples.desktop file2 hello Music new4 Publ
10 Documents file1 file3 hello1 new1.gz newnew5 rep1
ajouter Downloads file11 file55 log.txt new2.gz Pictures repta
user@ubuntu:~$ crontab -e
crontab: installing new crontab
user@ubuntu:~$ ls
: Desktop examples.desktop file2 hello Music new4 Publi
10 Documents file1 file3 hello1 new1.gz newnew5 rep1
ajouter Downloads file11 file55 log.txt new2.gz Pictures repta
user@ubuntu:~$ crontab -l

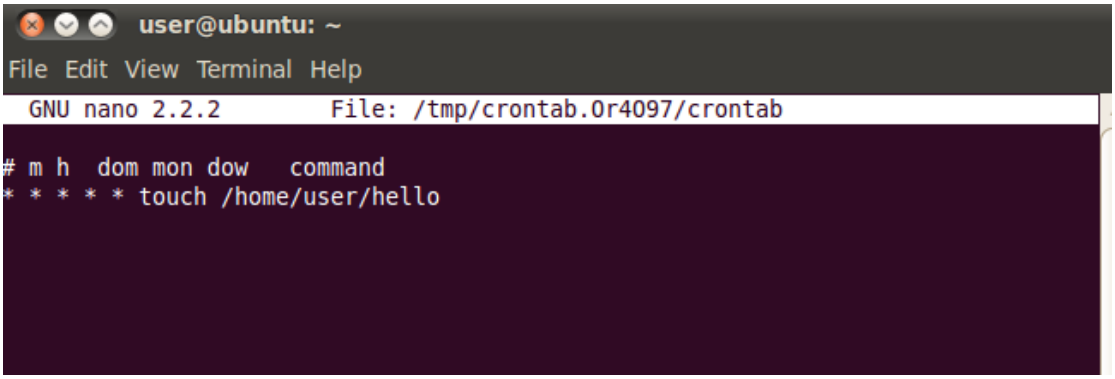
Affichage du fichier contenant les job:

>crontab –l

```
user@ubuntu:~$ crontab -l
no crontab for user
```

Edition du fichier crontab et choix de l'editeur nano:

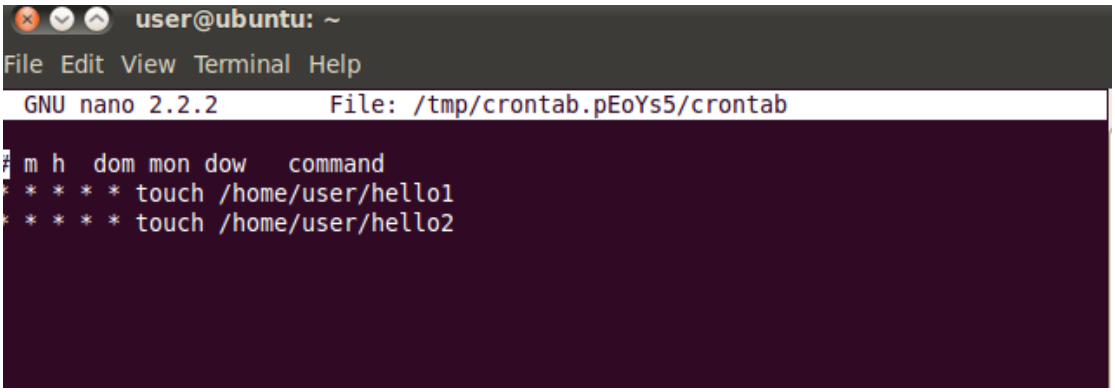
```
user@ubuntu:~$ crontab -e
```

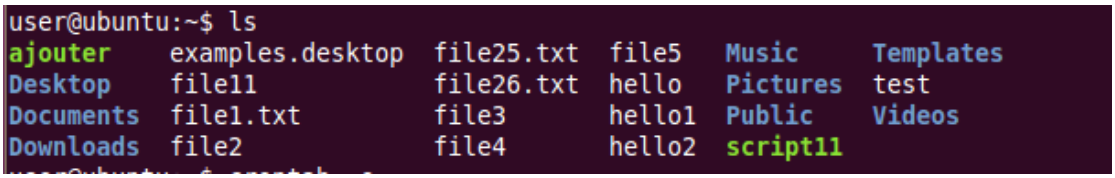


Consultation de l'heure en cours:

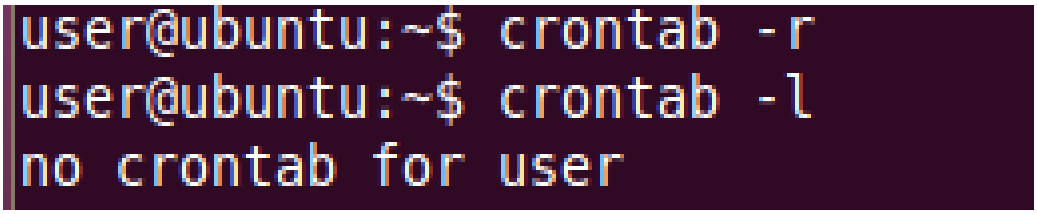
```
user@ubuntu:~$ date
Wed Apr 23 21:18:20 PDT 2025
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 21:19 hello
user@ubuntu:~$ ls -l hello
-rw-r--r-- 1 user user 0 2025-04-23 21:20 hello
```

Planifier deux taches:



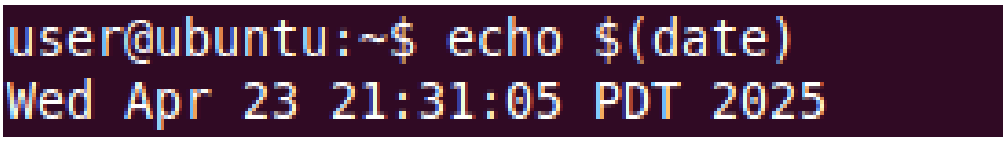


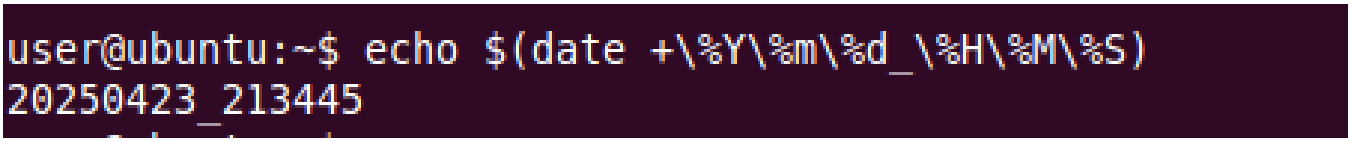
Supprimer toutes les planifications sur crontab:



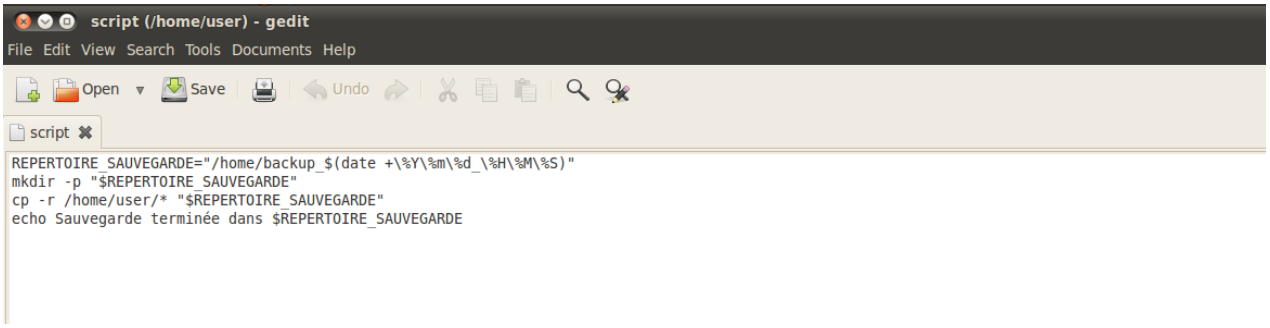
Faire un backup:

Avant de commencer:





Contenu du script:




```
user@ubuntu:~$ chmod u+x script
chmod: changing permissions of `script': Operation not permitted
user@ubuntu:~$ sudo chmod u+x script
user@ubuntu:~$ ls -l script
-rwxr--r-- 1 root root 198 2025-04-23 21:48 script
```

Lancer le script sans crontab:

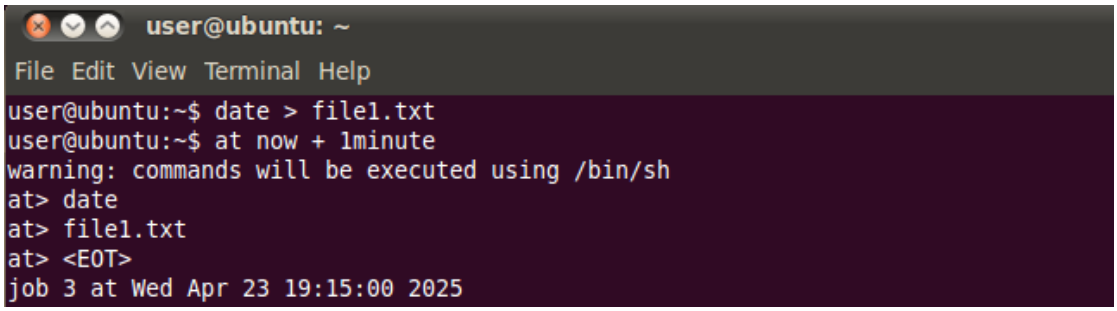
```
user@ubuntu:~$ ls -l /home
total 20
drwxr-xr-x  2 jack      jack      4096 2024-04-04 17:05 jack
drwxr-xr-x  2 jaqueline jaqueline 4096 2024-04-04 17:05 jaqueline
drwxr-xr-x 25 user      user      4096 2025-04-23 21:48 user
drwxr-xr-x  2 user3     user3     4096 2024-04-04 15:40 user3
drwxr-xr-x  2 user8     user8     4096 2024-04-04 16:25 user8
```

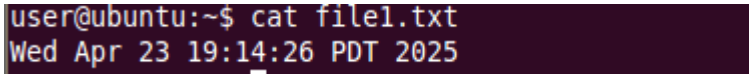
Lancer le script chaque minute:

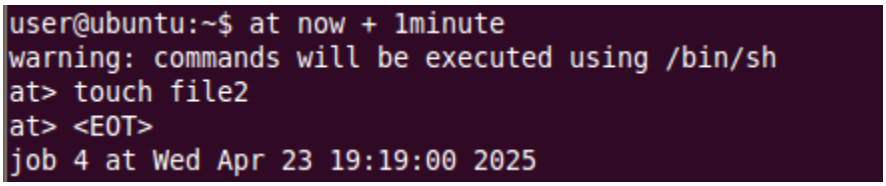
```
user@ubuntu:~$ cat ./log.txt
Sauvegarde terminée dans /home/backup_20250423_215501
Sauvegarde terminée dans /home/backup_20250423_215601
Sauvegarde terminée dans /home/backup_20250423_215702
```

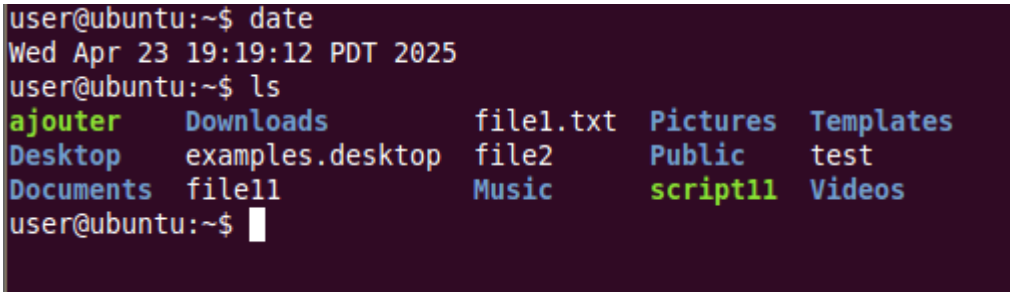
TP:
PLANIFICATION PONCTUELLE ET RECURRENTE

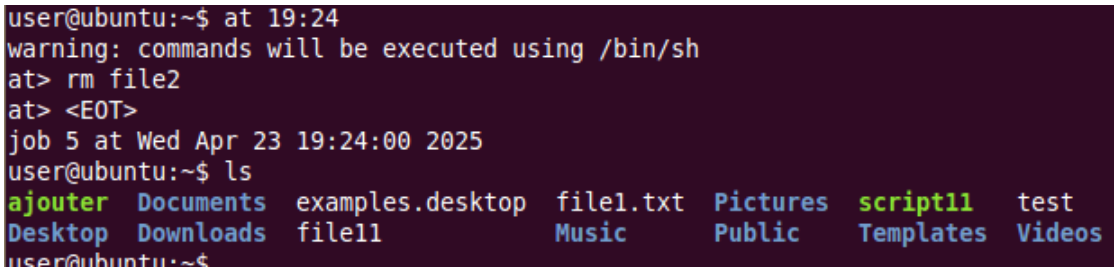
Executer une commande à une heure precise :











Plusieurs taches en une seule:

```
user@ubuntu:~$ at 19:30
warning: commands will be executed using /bin/sh
at> touch file2
at> touch file3
at> <EOT>
job 6 at Wed Apr 23 19:30:00 2025
user@ubuntu:~$
```

```
user@ubuntu:~$ at 19:30
warning: commands will be executed using /bin/sh
at> touch file2
at> touch file3
at> <EOT>
job 6 at Wed Apr 23 19:30:00 2025
user@ubuntu:~$ ls
ajouter    Downloads         file1.txt  Music     script11   Videos
Desktop    examples.desktop  file2      Pictures  Templates
Documents  file11            file3      Public    test
```

Avec echo et consultation des taches “>*atq*”:

```
user@ubuntu:~$ date
Wed Apr 23 19:41:20 PDT 2025
user@ubuntu:~$ echo "touch file4" | at 19:44
warning: commands will be executed using /bin/sh
job 7 at Wed Apr 23 19:44:00 2025
user@ubuntu:~$ atq
1       Thu Apr 24 00:35:00 2025 a user
7       Wed Apr 23 19:44:00 2025 a user
```

Voir le contenu d’une tache:

```
user@ubuntu:~$ echo "touch file5" | at 19:50
warning: commands will be executed using /bin/sh
job 8 at Wed Apr 23 19:50:00 2025
user@ubuntu:~$ sudo at -c 8
#!/bin/sh
# atrun uid=1000 gid=1000
# mail user 0
umask 22
ORBIT_SOCKETDIR=/tmp/orbit-user; export ORBIT_SOCKETDIR
SSH_AGENT_PID=1944; export SSH_AGENT_PID
XDG_SESSION_COOKIE=4267cfdd30e1765e84a289db563a603a-1745459793.388122-1246277795
; export XDG_SESSION_COOKIE
WINDOWID=58720259; export WINDOWID
GNOME_KEYRING_CONTROL=/tmp/keyring-luAVsw; export GNOME_KEYRING_CONTROL
GTK_MODULES=canberra-gtk-module; export GTK_MODULES
USER=user; export USER
LS_COLORS=rs=0:di=01\;34:ln=01\;36:hl=44\;37:pi=40\;33:so=01\;35:do=01\;35:bd=40
\;33\;01:cd=40\;33\;01:or=40\;31\;01:su=37\;41:sg=30\;43:ca=30\;41:tw=30\;42:ow=
```

```
PWD=/home/user; export PWD
GDM_KEYBOARD_LAYOUT=us; export GDM_KEYBOARD_LAYOUT
LANG=en_US.utf8; export LANG
GNOME_KEYRING_PID=1892; export GNOME_KEYRING_PID
MANDATORY_PATH=/usr/share/gconf/gnome.mandatory.path; export MANDATORY_PATH
GDM_LANG=en_US.utf8; export GDM_LANG
GDMSESSION=gnome; export GDMSESSION
SPEECHD_PORT=7560; export SPEECHD_PORT
SHLVL=1; export SHLVL
HOME=/home/user; export HOME
GNOME_DESKTOP_SESSION_ID=this-is-deprecated; export GNOME_DESKTOP_SESSION_ID
LOGNAME=user; export LOGNAME
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share/:/usr/share/; export XDG_DATA_DI
RS
DBUS_SESSION_BUS_ADDRESS=unix:abstract=/tmp/dbus-780GtnwBBW,guid=145728575974b28
bac4184fb68099a51; export DBUS_SESSION_BUS_ADDRESS
LESSOPEN=\|\ /usr/bin/lesspipe\ %s; export LESSOPEN
LESSCLOSE=/usr/bin/lesspipe\ %s\ %s; export LESSCLOSE
XAUTHORITY=/var/run/gdm/auth-for-user-TGLZQ7/database; export XAUTHORITY
COLORTERM=gnome-terminal; export COLORTERM
cd /home/user || {
         echo 'Execution directory inaccessible' >&2
         exit 1
}
touch file5
```

Execution à une date précise:

```
user@ubuntu:~$ echo "touch  file" | at 20:06 april 23 2025
warning: commands will be executed using /bin/sh
job 10 at Wed Apr 23 20:06:00 2025
user@ubuntu:~$ atq
1       Thu Apr 24 00:35:00 2025 a user
9       Thu Apr 24 10:00:00 2025 a user
10      Wed Apr 23 20:06:00 2025 a user
```

Supprimer un job:

```
user@ubuntu:~$ atq
1       Thu Apr 24 00:35:00 2025 a user
9       Thu Apr 24 10:00:00 2025 a user
10      Wed Apr 23 20:06:00 2025 a user
user@ubuntu:~$
user@ubuntu:~$
user@ubuntu:~$
user@ubuntu:~$ atrm 10
user@ubuntu:~$ atq
1       Thu Apr 24 00:35:00 2025 a user
9       Thu Apr 24 10:00:00 2025 a user
user@ubuntu:~$
```

Sleep:

```
user@ubuntu:~$ touch ss ; sleep 10 ; rm ss
```

Sleep 60: Pause pendant 60 secondes

```
user@ubuntu:~$ echo "touch ~/file25.txt && sleep 60 && touch ~/file26.txt" | at 20:15 Apr 23 2025
warning: commands will be executed using /bin/sh
job 11 at Wed Apr 23 20:15:00 2025
user@ubuntu:~$ ls
ajouter    Downloads         file1.txt   file3  Music     script11   Videos
Desktop    examples.desktop  file2       file4  Pictures  Templates
Documents  file11            file25.txt  file5  Public    test
user@ubuntu:~$ ls
ajouter    Downloads         file1.txt   file26.txt  file5     Public     test
Desktop    examples.desktop  file2       file3       Music     script11   Videos
Documents  file11            file25.txt  file4       Pictures  Templates
```